



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 17th August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : **EEE G539 (3 2 5)**

Course Title : **Modelling and Analysis of Electrical Machines**

Instructor-in-charge : **Prof. Aditya R. Gautam** (email id:aditya.gautam@pilani.bits-pilani.ac.in)

Practical/Lab Instructor : **Vedant Vyas**

- 1. Course description:** Review of Rotating Electrical Machines, Element of Generalized Theory of Rotating Electrical Machines, Linear transformation and Reference-frame Theory, DC Machines, Three-phase Synchronous Machines, Three-phase Induction Machines, Special Machines, Introduction to motor parameters estimation and model generation using MATLAB
- 2. Scope and Objective:** The objective of the course is to comprehend and analyse the steady-state, transient and dynamic behaviour of electrical machines using mathematical models, and also to familiarize the students about advanced modelling. The course will develop students' understanding on modelling of electrical machines from conventional methods to advanced methods through generalized theory of electrical machines. Furthermore, the students will appreciate the scope and applicability of course with the help of hands-on practice using simulation tools and hardware setup, encouraging and strengthening their ability to approach and investigate important challenges of automation and e-mobility associated to research interests of industries and academia, for example, digital-twin, model predictive modelling and control, fault analysis, predictive maintenance, and advanced control. The prerequisite of this course is, to have a basic understanding of electrical circuits, electrical machines and control systems; mainly modelling of physical systems.
- 3. Course learning outcomes:** Students will be able to,
 - a. Develop the model of electrical machines using actual machine variables and parameters.
 - b. Develop generalized theory (unified approach) based model of electrical machines.
 - c. Apply different transformations using reference frame theory to simplify machine modelling.
 - d. Develop the block-diagram of electrical machines using their dynamic model.
 - e. Develop and visualize machine-model using software such as MATLAB-Simulink, and use the model for analysis and control-loop design for hardware testing and implementation using real-time simulators such as TMS320F28379d.

4. Text books and references:

Text Books:

TB1: P. S. Bimbhra, Generalized Theory of Electrical Machines, Khanna Publishing House, 2021

TB2: Krause, P., Wasynczuk, O., Sudhoff, S., & Pekarek, S. Analysis of electric machinery and drive systems. 3rd edition, Wiley-IEEE Press.

Reference Books:

RB1: Krishnan, R., Electric Motor Drives: Modeling, Analysis, and Control. Prentice Hall, 2024

RB2: Ong, C. M., Dynamic Simulation of Electric Machinery: Using MATLAB/SIMULINK. Prentice Hall, 1998.

RB3: Sang-Hoon Kim, Electric Motor Control: DC, AC, and BLDC Motors, Elsevier Science.

5. **Method of conduct of the course:** Teaching Board, PPTs, Seminars and Experimental kits.

6. **Course Plan:** Three lectures (each of 50 minutes) and a Lab of two hours weekly.

Module, Lectures, (Total Lecture)	Learning objectives	Topics to be Covered	Reference
M-1 1-6 (6L)	Review of Rotating Electrical Machines	- Recap of electrical machines and basic principles of analysis - Magnetically coupled circuits - Machine windings and air-gap MMF - Winding inductance and voltage equations	TB2: Ch-1 RB3: Ch-4
M-2 7-10 (4L)	Element of Generalized Theory of Rotating Electrical Machines	- Essentials of rotating machines - Sign-conventions - Basic two-pole machine - Per unit system - Kron's primitive machine - Analysis of electrical machines	TB1: 1-1.4, 1.6-1.8
M-3 11-14 (4L)	Linear transformation and Reference-frame Theory	- Invariance of power - Transformations in Electrical Machines - Commonly used reference frame and transformation between them.	TB1: 2.1-2.9, TB2: 3.1-3.7, RB3: Ch-4
M-4 15-20 (6L)	DC Machines	Conventional modelling using system variables, state equations, transfer functions, block-diagram	TB1: 3.2-3.6, 3.7 TB2: 2.5-2.7



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		- Generalized theory based modelling of separately excited dc generator and motor, series and shunt motor - Linearization techniques	RB1: 2.5-2.6
M-5 21-28 (8L)	Three-phase Synchronous Machines	- Machine parameters, voltage/torque equations, and their reference frame variables - Reactance and time constants - Steady-state, Short-circuit, transients and dynamic analysis - Dynamic performance at torque change and three-phase fault	TB1: 4.1-4.9, 4.11-4.18 TB2: 5-1-5.11
M-6 29-36 (8L)	Three-phase Induction Machines	- Equivalent circuit, Voltage and torque equations, - Transformation, reference frame variables - Steady-state analysis - Dynamic performance analysis at torque change and three-phase fault	TB1: 5.1-5.5, 5.7 TB2: 4.1-4.6, 4.8-4.9, 4.12-4.14 RB1: 5.4-5.5, 5.7, 5.9-5.11
M-7 37-42 (6L)	Special Machines	- Trapezoidal and sinusoidal BLDC - Basics of VRM and PMSM - Model of PMSM and BLDC - Basics of VSI drives steady-state and average values modelling and analysis	TB1: 9.2, 9.3.1-9.3.5, 9.4, TB2: 15.1-15.6, 15.12, RB1: 9.4.1, 9.10.1-9.10.2

7. List of Experiments

1. Study on operating principles electrical machines with the help of MATLAB-Simulink models and Experimental setup.
2. Extraction of transformed components (Park and Clarke transforms using reference frame theory) of three-phase supply with the help of TMS F28379D development kit.
3. To study the use of SOGI filters for extraction of α - β components and harmonics in single-phase supply using MATLAB-Simulink/experimentation.
4. Modelling of dc machine using MATLAB Simulink, and auto tuning of PID controller.
5. Slip test: To calculate direct and quadrature reactances of the synchronous machine.
6. Modelling of three-phase induction motor using PSIM/MATLAB-Simulink.
7. Modelling of three phase synchronous motor using PSIM/MATLAB-Simulink.
8. Modelling and block-diagram development of control-structure of BLDC using Matlab-simulink.
9. Modelling and block-diagram development of control-structure of PMSM using Matlab-simulink.
10. Fault Simulation and Diagnosis in Electrical Machines using MatLab-simulink

8. Evaluation Scheme:

(Given below are suggestions for possible kinds of evaluation components. Indicative only.)

Component	Weightage (100 M)	Duration	Date and time	Comments
Lab	[20%]	Lab hours	TBA	Lab Quiz/Experiment/Simulation
Homework Assignments	[20%]	one week	TBA	Take home/Seminar
Mid Semester	[25%]	90 min	TBA	Closed-Book (CB)
Comprehensive	[35%]	180 min	TBA	Open-book (OB)-CB

Criterion for NC: A student should obtain 25% of the average marks of the Class, or 10% of the Course Total Marks, whichever is lower to clear the course. If any student gets the marks lower than the prescribed standard mentioned above, he/she may be awarded NC Report.

9. Chamber consultation hours: To be announced in lecture hour.

10. Makeup Policy: Makeup will be granted to extremely genuine cases only.

11. Notices: All notices will be displayed on Nalanda only (<https://nalanda-aws.bits-pilani.ac.in/>)

Instructor-in-charge: Aditya R. Gautam

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